Does Spending More per Student Mean More College Graduates? A Look at Maryland's Counties

(Matthew Velasquez) DATA 110: Data Visualization and Communication · Project 1 · October 2026

Introduction

I wanted to find out if counties that spend more per student in Maryland also tend to have more adults with a college degree. To do this, I utilized the Choose Maryland "Compare Counties: Education" data as well as three visuals. I found that the relationship between spending and the number of adults with a college degree is essentially none ($r = 0.07$).

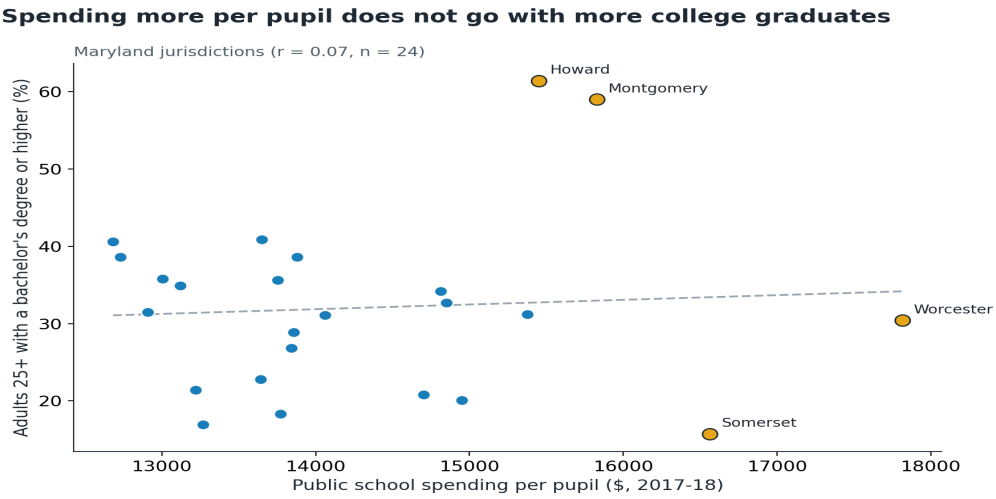
The Data

There are 24 observations (23 counties and Baltimore City) and 10 variables in the dataset. Variables include spending per student, student-teacher ratio, and the percentage of adults with a bachelor's degree. Data sources include the Maryland State Department of Education, the U.S. Census Bureau, and the Maryland Higher Education Commission (2014-2019).

Spending per student varies between $12,682 and $17,814. The percentage of adults with a bachelor's degree varies from 15.7% (Somerset) to 61.4% (Howard), with a median of 31.4%.

The only blanks occur in the columns for college enrollment. A blank indicates that the county does not have a college in that category. The exception to this is Dorchester, which has a 4-year college but no enrollment data. Enrollment data was not used in any of the charts.

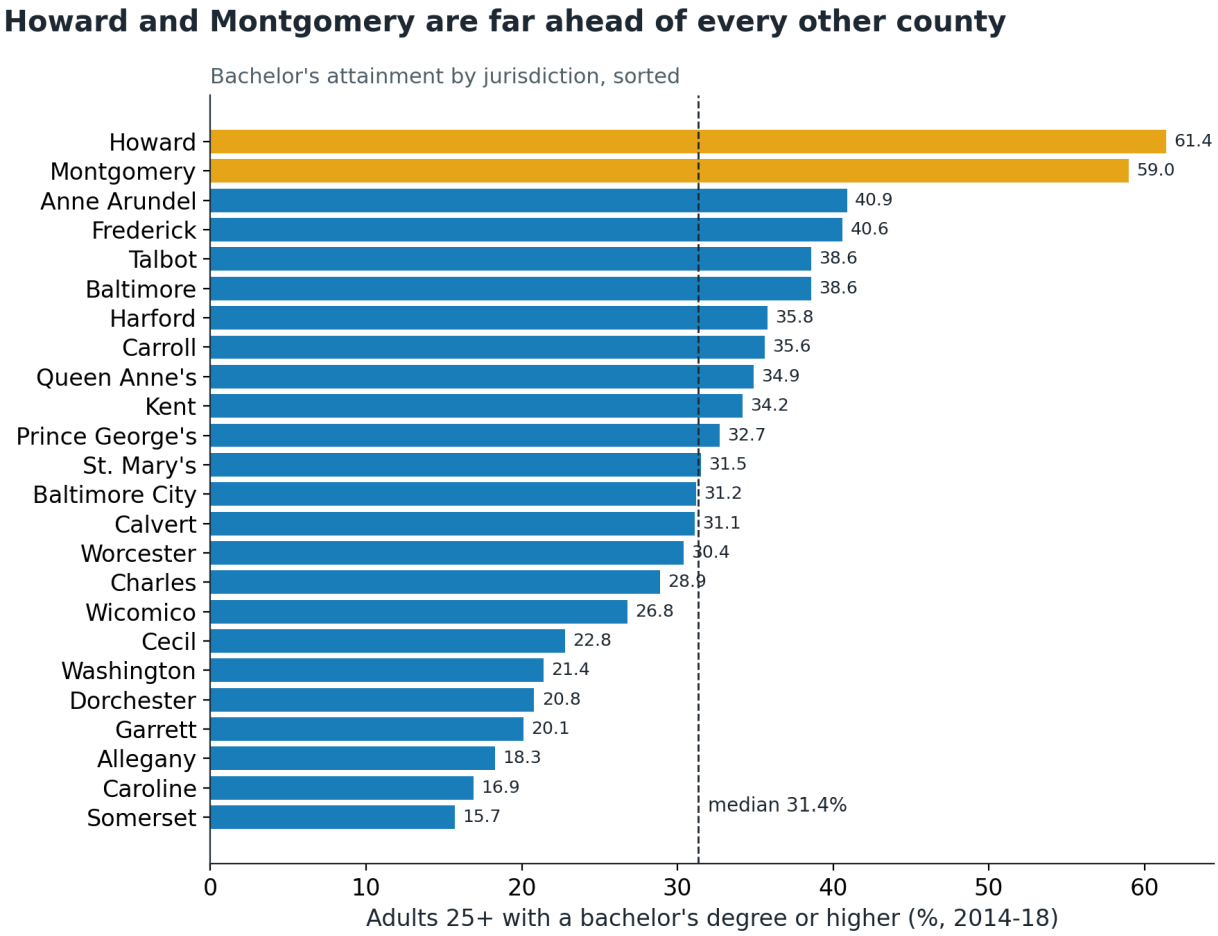
Chart 1: Spending and Degrees Are Almost Unrelated



Caption: These dots represent counties. There is no correlation between dollars spent per student and the percent of adults with a bachelor's degree. The correlation for the dashed line is nearly zero (r = 0.07).

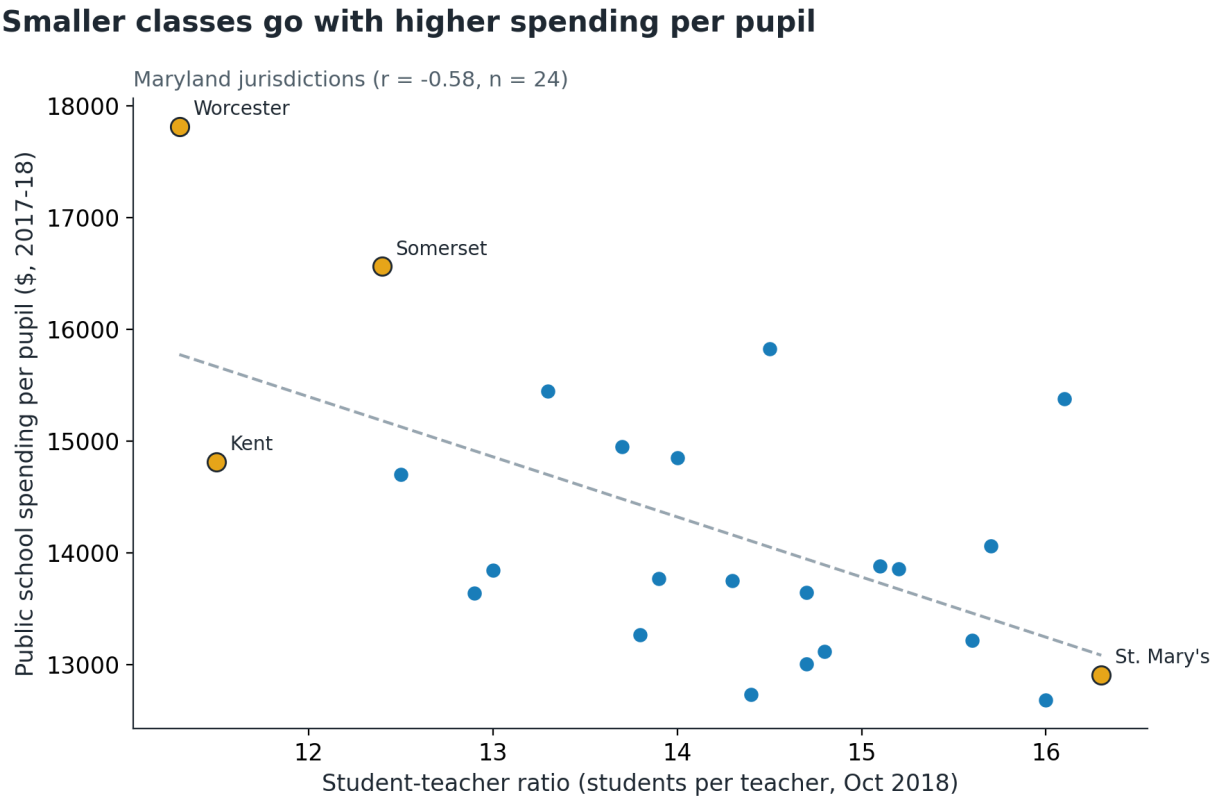
Somerset spends $16,563 per student and has 15.7% bachelor's attainment. Howard spends $15,449 and has 61.4%.

Chart 2: Two Counties Are Far Ahead



Caption: Bachelor's attainment in each county from highest to lowest. Howard and Montgomery are shown by gold bars. The dashed line represents the median (31.4%). Howard (61.4%) and Montgomery (59.0%) are approximately 18 percentage points higher than the following county, Anne Arundel (40.9%). This difference is not explained by spending.

Chart 3: Spending Follows Class Size



Caption: Each dot represents a county. On the x-axis there are fewer students per teacher, and on the y-axis more money per student. The dashed line has a correlation of r = -0.58.

Worcester has the lowest number of students per teacher (11.3) and the highest money per student ($17,814). This only signifies a relationship, not causation.

Conclusion

More spending per student is not associated with higher degrees (r = 0.07). However, spending is associated with smaller class sizes (r = -0.58). Howard and Montgomery have higher degree rates compared to all other counties.

There are two caveats. First, spending data reflect current students, while degrees reflect adults 25 and older who attended school in the past. Second, there are only 24 counties, and correlations are highly sensitive to the presence or absence of one or two outliers.